

A PROJECT TITLE
ON
Wine Quality Classification Using Logistic Regression and Multinomial Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF CSE



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Name of Title: Wine Quality Classification Using Logistic Regression and Multinomial Logistic Regression

1. Abstract:

Wine quality assessment is traditionally performed by human tasters through sensory evaluation, a process that is subjective, time-consuming, and prone to inconsistency. This project addresses these limitations by developing a machine learning based system that predicts wine quality using physicochemical properties such as acidity, residual sugar, chlorides, sulphur dioxide levels, density, pH, sulphates, and alcohol content. The dataset used is the publicly available Wine Quality dataset, which contains laboratory measured attributes of red and white wine samples along with quality ratings assigned by expert tasters.

Two classification approaches are explored: binary Logistic Regression, which classifies wines as good or poor quality based on a defined threshold, and Multinomial Logistic Regression, which extends the model to predict multiple quality classes directly, capturing finer distinctions in quality scores. The workflow includes pre-processing steps such as handling missing values and feature scaling, exploratory analysis of correlations between chemical properties and quality, and an 80:20 train-test split. Both models are implemented in Python using scikit-learn, and performance is evaluated using accuracy, precision, recall, F1-score and confusion matrices.

The implementation relies on Python 3.x with the pandas, NumPy, scikit-learn, Matplotlib and Seaborn libraries, developed and executed within Jupyter Notebook or Google Colab. Results show that Multinomial Logistic Regression provides more granular predictions across the full range of quality scores, while binary Logistic Regression offers a simpler, more interpretable model for quick quality screening. Overall, the project demonstrates how machine learning can supplement subjective sensory panels with a data-driven, reproducible and scalable approach to wine quality assessment, with practical value for quality control in the wine industry.

2. INTRODUCTION:

Wine production is a global industry in which quality directly influences market value, brand reputation and consumer trust. Traditionally, wine quality is certified through sensory evaluation carried out by panels of expert tasters who judge attributes such as aroma, taste and balance. While effective, this approach is inherently subjective, expensive to scale, and susceptible to variation between tasters and even between sessions with the same taster. With the growing availability of physicochemical laboratory data collected during production, such as fixed and volatile acidity, citric acid, residual sugar, chlorides, free and total sulphur dioxide, density, pH, sulphates and alcohol content, it becomes possible to model the relationship between measurable chemical composition and perceived quality using statistical learning techniques.

This project explores the use of Logistic Regression and its multinomial extension to classify wine samples based on quality. Logistic Regression is first used to build a simple binary classifier that separates wines into good and not-good categories, providing an interpretable baseline. The model is then extended to a Multinomial Logistic Regression setup capable of directly predicting the full range of discrete quality scores present in the dataset, allowing for a more detailed and realistic classification. By comparing the two approaches, the project aims to show how machine learning can automate and standardize wine quality assessment while remaining transparent and interpretable, making it suitable for practical use in winery quality-control pipelines.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10/11, macOS, or Linux
- Programming Language: Python 3.8 or above
- Development Environment: Jupyter Notebook / Google Colab / VS Code
- Libraries: NumPy, Pandas, scikit-learn, Matplotlib, Seaborn
- Package Manager: Anaconda / pip
- Dataset Source: UCI Machine Learning Repository - Wine Quality Dataset
- Version Control: Git and GitHub (for source management)



Signature of the Guide



Course Instructor